

FLEX-UF: Spatially Adaptive Early Exit in DCVC-UF

Anonymous CVPR submission

Paper ID DRAFT

Abstract

001 *The benefit of another decoder layer varies across an image,*
002 *yet DCVC-UF applies the same synthesis depth everywhere.*
003 *We introduce FLEX-UF, a spatial early-exit extension of its*
004 *intra model. A shared coded representation and synthesis*
005 *prefix feed tiles that stop at different depths. Exit adapters*
006 *assemble their features for one full-frame repair and recon-*
007 *struction head; only surviving tiles execute later blocks. We*
008 *evaluate two distinct sources of efficiency: reducing aver-*
009 *age depth and choosing where to retain it. Across 53 intra*
010 *frames at five quality indices, learned routing saves 27.97%*
011 *of modelled synthesis multiply-accumulates at a nominal*
012 *0.1 dB YCbCr-MSE target under per-source calibration. Of*
013 *this saving, 2.54 percentage points exceed ordered dithering,*
014 *which mixes depths without predicting their spatial value.*
015 *Retrospective selection under a common measured-loss cap*
016 *retains a 2.93-point margin over all 265 pairs. This margin*
017 *contracts as the shallowest exit saturates. Fixing controls on*
018 *other sequences removes per-source calibration, but a QP32*
019 *reconstruction replay does not yet resolve the paired routing*
020 *advantage. These results identify where adaptive depth can*
021 *add value while separating arithmetic savings from unmea-*
022 *sured codec latency. An ongoing independent-depth study*
023 *extends the same allocation question from shared exits to*
024 *codec selection.*

025 1. Introduction

026 Consider two regions decoded from the same image. In one,
027 another pair of synthesis blocks improves reconstruction;
028 in the other, it changes little. A fixed-depth decoder pays
029 for both. Efficient operators make this work cheaper, but
030 leave the allocation unchanged. Spatial early exit asks a
031 complementary question: *where is further decoding worth*
032 *its cost?*

033 We investigate this question in the intra model supplied
034 with DCVC-UF [7]. Its synthesis transform contains twelve
035 constant-width blocks between upsampling and reconstruc-
036 tion, providing a concrete axis along which to vary capacity.
037 We turn this fixed stack into an image-dependent allocation

of depth. Following the incremental-benefit view of adaptive
patch exiting [11], a useful decision reflects what additional
processing can recover, rather than texture alone. In a codec,
that benefit also depends on the information preserved at the
chosen rate: a visually complex region need not justify the
deepest reconstruction.

There are two natural ways to make this decision (Fig. 1).
Across models, a selector assigns each patch to a codec of
suitable capacity, following the patch-to-network principle
of ClassSR [6]. Within a model, different patches leave a
shared synthesis stack at different depths. The distinction is
where the commitment is made. An independent codec is
chosen before encoding and can learn its own representation;
an exit is chosen after one representation has been recovered
and reuses its synthesis prefix. FLEX-UF implements and
evaluates the latter. The six-depth bank is a complementary
study of how specialisation trades against this reuse.

Inside a codec, early exit requires more than attaching in-
termediate outputs. Features at different depths must support
a common reconstruction; conditional tiles lose some spatial
context; and choosing an exit can itself require expensive
candidate decoding. FLEX-UF addresses execution with
a full-frame stem, a shrinking batch of active tiles, point-
wise exit adapters, and a shared repair and reconstruction
head. Exited tiles leave the active batch, making the con-
ditional path an actual reduction in executed blocks. The
frozen analysis and entropy path supplies one common la-
tent; its payload can support different stopping maps without
re-encoding the image.

We separate the value of *cheaper depth* from the value of
content-dependent placement. Uniform depth tests global
truncation. Ordered dithering permits mixtures of depths
without predicting where extra synthesis is useful. Under
source-calibrated control, learned routing improves on this
stronger baseline by 2.54 percentage points of synthesis
MAC saving at a nominal 0.1 dB target, above dither-
ing's 25.43% reduction. That increment shrinks near the
shallowest-exit ceiling. Adaptation has most room to help
when extra depth still matters but does not matter equally
everywhere. A common delivered-loss analysis retains this
pattern. A separate QP32 replay fixes controls on other

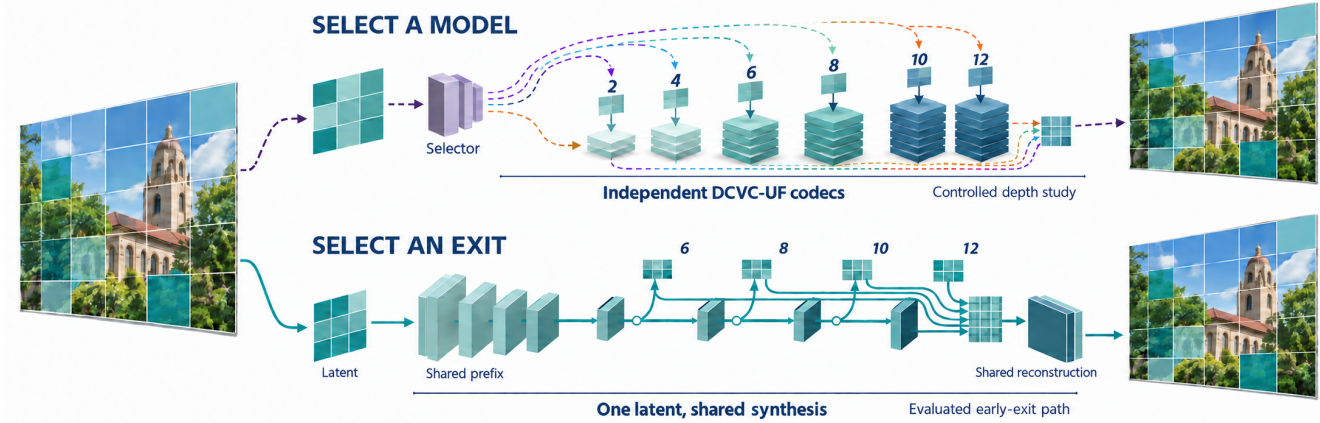


Figure 1. **One allocation principle, two ways to vary synthesis depth.** *Top:* select a complete DCVC-UF codec before encoding a patch; the six-depth bank and its selector form the controlled, prospective study. *Bottom:* select a stopping depth within one shared-latent decoder; this is the evaluated early-exit path. The paths are alternatives, not successive stages. AI-generated conceptual illustration: picture planes are illustrative, not dataset samples or reconstructed outputs. Exact execution and measured allocations appear in Figs. 2 and 6.

sequences; its paired routing margin remains statistically unresolved. This separates evidence for useful allocation from evidence that a source-independent stopping control reliably captures that value.

We contribute a spatially conditional DCVC-UF synthesis path and a controlled evaluation of learned placement. Uniform depth, blind mixtures, a common measured-loss cap and fixed-control replay distinguish the architecture’s saving from the selector’s value. Frozen-weight interventions and a stream-only decoding check probe the mechanism. The independent-depth study extends this allocation question to codec selection; complete codec speedups remain a question for measured execution.

2. Related Work

Making learned decoding cheaper. Shallow synthesis transforms reduce decoder computation globally; stronger analysis or iterative encoding can compensate for some of the lost reconstruction capacity [14]. DCVC-RT instead emphasises operational costs such as memory traffic and function calls [4], while DCVC-UF increases video throughput with chunk-based coding and streamlined entropy interaction [7]. We modify only DCVC-UF’s intra synthesis transform. These approaches motivate two controls in our study: a fixed shallow decoder isolates the value of adaptation, and full execution measurements are required before arithmetic reduction can be called acceleration.

Choosing a network or choosing an exit. ClassSR learns to assign super-resolution patches to networks of different capacity [6]. APE instead lets patches leave a multi-exit network according to predicted incremental reconstruction benefit [11]; it already shares the reconstruction tail across exits.

AdaRevD predicts patch difficulty to choose a deblurring sub-decoder [8]; RBQE terminates compressed-image enhancement early [12]. Swift introduces compute-dependent early exits in a layered neural video decoder, with different output resolutions [2]. FLEX-UF addresses the codec-specific case within DCVC-UF synthesis at fixed output resolution: one entropy-decoded latent feeds conditional feature tiles, exit-specific adapters align those features, and reconstruction occurs after full-frame assembly. Our proposed independent-depth bank follows the ClassSR allocation principle, but selecting a codec also selects an analysis transform and entropy model.

Adaptive capacity inside a codec. Slimmable compressive autoencoders jointly optimise rate and distortion at multiple network widths [13]. AdaNIC is the closest spatial-capacity precedent: blockwise spatially adaptive operators and a router vary transform width within an image [10]. For video, cgSlimDecoder learns channel allocations across decoder modules under complexity constraints [3]. FLEX-UF varies *depth* through nested, constant-width synthesis blocks and executes later stages on a shrinking tile batch. Its evaluation asks an additional question: how much of the saving requires content-aware placement after blind mixtures have already reduced the average depth?

Spatial selection among complete codecs. Blard et al. train content-specialised codecs of the same architecture, search their regional rate–distortion costs, and signal a mode map [1]. Their method also processes connected regions of one mode jointly. Our region-merging diagnostic therefore tests an existing strategy in DCVC-UF, rather than introducing it. The planned bank varies synthesis depth and inves-

141 tigrates selection before candidate encoding; the evaluated
142 shared-exit path instead retains one representation. These
143 distinctions separate specialisation, capacity and representa-
144 tion reuse, which cannot be inferred from a comparison of
145 model sizes alone.

146 **Preserving context across blocks.** Kim et al. derive over-
147 lap and boundary rules that preserve full-image CNN recon-
148 struction [5]. FLEX-UF instead accepts context loss in its
149 zero-halo suffix and learns a common repair/head. Its tiled
150 all-deep output must therefore be evaluated separately from
151 full-frame synthesis.

152 3. Spatially Adaptive DCVC-UF

153 3.1. The allocation problem

154 Let $\mathbf{a} = (a_1, \dots, a_N)$ assign a synthesis option to each re-
155 gion. We seek the least computation that keeps the assembled
156 reconstruction within a quality-loss budget:

$$157 \min_{\mathbf{a}} \mathcal{C}(\mathbf{a}) \quad \text{subject to} \quad \Delta(x, \hat{x}_{\mathbf{a}}; \hat{x}_{\text{ref}}) \leq \delta, \quad (1)$$

158 where $\mathcal{C}$ is the chosen compute cost and $\hat{x}_{\text{ref}}$ fixes the
159 reconstruction anchor. The main sweep uses $\Delta_{444} =$
160 $10 \log_{10}[D_{444}(x, \hat{x}_{\mathbf{a}})/D_{444}(x, \hat{x}_{\text{ref}})]$, with $D_{444} = \|x -$
161 $\hat{x}\|_2^2/(3HW)$ on equal-channel, unclipped YCbCr 4:4:4 ten-
162 sors. Positive values denote a quality loss in dB. The con-
163 straint belongs to the *assembled image*: spatial interactions
164 can invalidate an additive tile-error approximation.

165 For early exit, a_t identifies a stopping point after one
166 latent has been recovered. For the prospective model bank,
167 a_t identifies a complete codec before encoding, making both
168 payload rate and reconstruction depend on the assignment.
169 The bank must therefore also be compared at matched rate,
170 including expert signalling (Sec. 5). We implement shared
171 exits below.

172 3.2. Shared latent, variable depth

173 Let $\hat{y}$ be the recovered latent of the fixed DCVC-UF anal-
174 ysis transform and entropy model. The synthesis network
175 upsamples it by two and processes 384-channel features at
176 one eighth of the image resolution. Four of its twelve trunk
177 blocks form the full-frame stem; the remaining eight form
178 four conditional block pairs.

179 The stem output is partitioned into 32×32 feature tiles,
180 each covering 256×256 image pixels. For a tile assigned
181 depth k , let $f_t^{(k)}$ be its feature after the corresponding block
182 pair and let A_k be its adapter. The complete image is recon-
183 structed as

$$184 \hat{x}_{\mathbf{k}} = H\left(q \odot G\left[\text{stitch}_t A_{k_t}(f_t^{(k_t)})\right]\right), \quad (2)$$

185 where G is grid-conditioned repair, q is the decoder’s quality
186 scaling, and H is the common output head, including pixel

187 shuffle. Adaptation acts on features before the shared head,
188 so tiles with different stopping depths need not be stitched
189 directly in image space.

190 **Reachable depth and actual execution.** The reachable
191 depths are $\mathcal{K} = \{6, 8, 10, 12\}$; depths two and four are in-
192 side the shared stem and are not evaluated exits. At each
193 conditional stage the implementation gathers only the surviv-
194 ing tiles, executes the next block pair, and writes departing
195 tiles through their adapters. A deeper tile reuses all com-
196 putation already performed. Evaluating all four candidates
197 and masking their reconstructions would not implement this
198 execution saving. For the recorded router map at 0.1 dB, the
199 active population decreases from 40 to 27, seven and three
200 tiles (Fig. 3).

201 3.3. Exit alignment and boundary context

202 An intermediate feature is not automatically compatible with
203 the final head. The exit adapter supplies the missing trans-
204 formation with pointwise residual operators. Its residual
205 output is zero-initialised, preserving the inherited feature
206 at initialisation. Adapter capacity scales with the depth re-
207 placed. Pointwise operators avoid adding an exit-specific
208 spatial receptive field; their arithmetic is included in the cost
209 model. The deepest exit uses an identity connection rather
210 than an additional learned adapter.

211 Independent tile execution still removes context from the
212 trunk’s spatial operators. The conditional trunk uses replicate
213 padding without a halo. After stitching, a depthwise spatial
214 filter and pointwise mixing act on the full feature canvas.
215 A learned gate indexed by position within the tile gives the
216 repair access to the known boundary lattice. The head then
217 operates on the complete canvas. This design places context
218 recovery at the output stage rather than expanding every
219 conditional block with an overlapping halo. We measure
220 the trained checkpoint’s dependence on adapters and repair
221 in Sec. 4.7; matched retraining is needed to choose their
222 capacity.

223 3.4. Learning exits and their selector

224 The evaluated shared-latent checkpoint, denoted e15, is ini-
225 tialised from released DCVC-UF weights. Analysis and
226 entropy parameters remain frozen; the synthesis network,
227 adapters and repair are fine-tuned. Training samples tile
228 depths and applies reconstruction supervision through the
229 mixed tiled path. Thus the adapters learn under the same
230 stitching and full-frame head used at inference. The in-
231 spected training configuration also enables a full-depth re-
232 construction anchor against the frozen release and adjacent-
233 exit feature distillation. Its objective can be written as

$$234 \mathcal{L} = \mathbb{E}_{\mathbf{k}}[R + \lambda D(x, \hat{x}_{\mathbf{k}})] + a\mathcal{L}_{\text{anchor}} + b\mathcal{L}_{\text{feature}}. \quad (3)$$

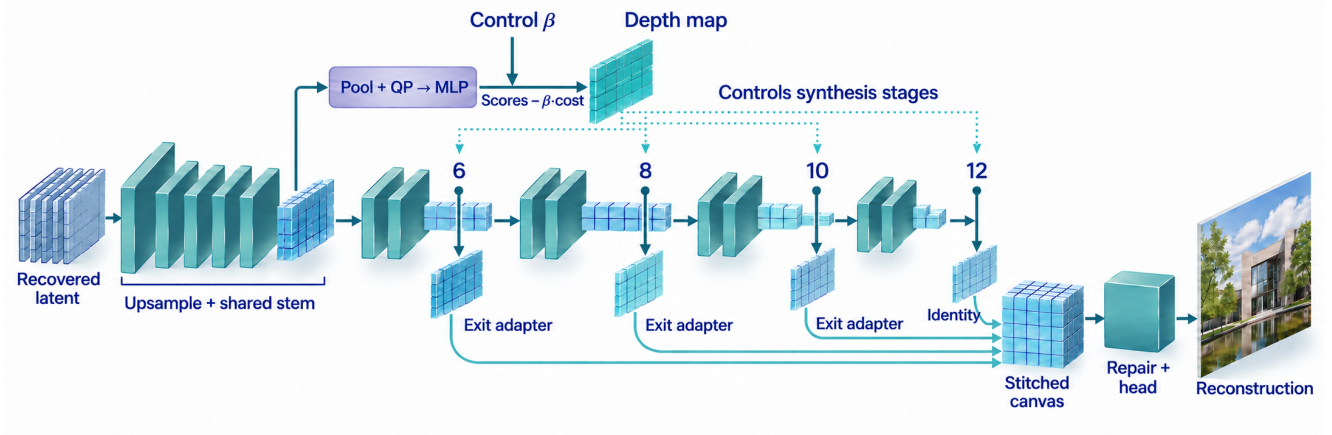


Figure 2. **Share the representation; spend further depth only on surviving tiles.** A four-block shared stem supplies the MLP with features and QP. Cost-adjusted scores select exits at total depths 6/8/10/12. Each departing feature passes through its adapter (identity at depth 12) into one canvas, followed by full-frame repair and reconstruction. The illustration follows the implemented connectivity; tensor contents and the output scene are conceptual, AI-generated artwork rather than measured reconstructions.

Only surviving tiles run the next block pair

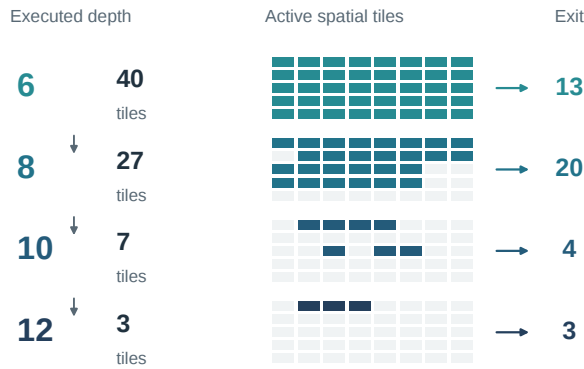


Figure 3. **A spatial decision becomes a smaller workload.** The recorded videoSRC05 map at QP32 leaves fewer tiles active at each successive block pair. Coloured cells run the indicated stage at their padded spatial coordinates; pale cells have already exited. Departures feed the aligned canvas in Fig. 2. The suffix evaluates 77 tile pairs versus 160 for an all-deep map; this work count does not measure GPU time.

tile errors $M_{t,k}$ and normalised costs c_k , its soft training labels favour low values of $M_{t,k} + \lambda_c c_k$. Soft-label cross-entropy and a regret term penalise costly decisions while allowing nearly equivalent exits to receive similar scores. The checkpoint uses $\lambda_c = 1.3 \times 10^{-5}$ and temperature ratio 0.1; the exact target, input layout and optimisation settings are given in the supplement. This objective learns a preference among exits, not a bound on final image error.

3.5. Stopping under a quality target

For equal padded tiles, the arithmetic model is

$$C(\mathbf{k}) = \frac{1}{N} \sum_t c_{k_t}, \quad S(\mathbf{k}) = 100[1 - C(\mathbf{k})]. \quad (4)$$

Shared stem and output work, exit adapters and conditional depth are included in c_k . Entropy-recovery networks, arithmetic coding, router execution, signalling and system overhead are outside this synthesis MAC quantity. The four costs are approximately $\{0.609, 0.758, 0.870, 1.010\}$ relative to full-frame synthesis. In particular, a tiled all-deep map can cost more than the full-frame reference because repair is still executed.

Source-informed search minimises $M_{t,k} + \lambda c_k$ per tile and adjusts λ for a target table loss. The learned alternative uses

$$k_t(\beta) = \arg \max_k \{\ell_{t,k} - \beta c_k\}, \quad (5)$$

where ℓ denotes log scores. Uniform depth selects one exit per frame. Bayer dithering mixes adjacent exits with a fixed spatial ordering. Its mixture fraction is calibrated under the same target, so it controls for finer average depth without predicting spatial difficulty.

In the recorded sweep, all controls are calibrated per source frame using stored error tables. This measures the

Here the latent-rate term R is shared across exits. With the front end frozen it does not optimise exit-dependent rate. The archived fine-tuning configuration uses 512-pixel crops and batch size eight; it is distinct from the new full 105-epoch official-recipe depth study in Sec. 5. The supplement records the available training provenance and its limits.

The stopping predictor projects the stem from 384 to 48 channels, pools each tile’s mean and standard deviation, and incorporates normalised QP. An MLP with two 256-unit hidden layers produces exit log scores. For candidate

allocation available with encoder-side source access. The fixed-control experiment in Sec. 4.6 removes per-source calibration. Increasing β cannot increase selected cost, but source distortion need not be monotone; the supplement audits the finite assignment path for both router and dither.

3.6. Quality references and decision information

The table target compares padded YCbCr 4:4:4 candidate errors to a released-weight full-depth reference. Final reported loss instead compares the cropped mixed output to the cropped, fine-tuned e15 full-frame output. These anchors are not interchangeable: the encoder/entropy weights match, while inherited synthesis weights have changed. Full-frame repair also couples neighbouring tiles. A table threshold therefore does not guarantee the same numerical loss after final reconstruction. We retain this distinction in the nominal sweep and add a retrospective common-reference delivered-loss comparison. A separate QP32 replay on all 53 frames measures the e15 full-frame RGB loss against released synthesis as 0.00908 dB; details are in the supplement.

An encoder-selected map must be signalled to the decoder; alternatively, a decoder-local predictor must recover the same decisions from available features and a reproducible control. We use explicit two-bit exit indices in a CPU stream-only check (Sec. 4.7). Its common payload confirms the representation-sharing path, while headers and map bytes remain part of the complete rate. Source access is legitimate at the encoder; its candidate evaluations and calibration must be charged to the decision cost.

4. Experiments

4.1. Evaluation protocol

We evaluate the first intra frame of 53 video sequences at QP values $\{0, 16, 32, 48, 63\}$, giving 265 frame-QP pairs. Every policy uses the same pinned e15 shared-latent checkpoint, the same tile geometry and the same per-exit cost vector. The learned rule uses one fixed router checkpoint. Six target budgets span 0.05–0.5 dB. The source-calibrated archive contains actual mixed reconstructions. Additional CPU experiments test fixed controls, the released reference and component dependence. Every numerical decoder comparison concerns DCVC-UF; no experiment here measures complete GPU codec latency.

Two complementary protocols answer different questions. The *nominal sweep* compares maps selected under the same table target, retaining the common feasible cases of all four rules: 204 pairs at 0.05 dB, 263 at 0.1 dB and 265 at each larger target. The *delivered-cap analysis* reselects from the outputs already measured at those targets under a common cropped YCbCr 4:4:4-loss cap, with the full-frame e15 output as an explicit fallback. It retains all 265 pairs at every cap. The second protocol is retrospective and source-aware;

its cost omits acquiring and rejecting candidates.

Confidence intervals use 5,000 paired bootstrap draws over sequences, keeping QPs from the same sequence together. They quantify sequence-sample variation conditional on fixed weights and controls, without including training or calibration uncertainty. The supplement also reports a conservative content-group sensitivity check. A separate fixed-cohort comparison retains the same 204 pairs across the entire nominal sweep, avoiding changes of sample membership as the target relaxes.

4.2. The reconstruction value of depth

The stored uniform-exit reconstructions provide a direct depth control. Across all 265 pairs, their mean padded YCbCr 4:4:4 losses against the released full-frame reference are 0.205/0.097/0.061/0.038 dB at 6/8/10/12 blocks. Their modelled MAC savings are 39.12/24.22/13.03/−0.95%, respectively. Additional depth has diminishing average benefit and non-uniform local benefit: mean within-frame tile-gain IQRs are 0.075/0.034/0.024 dB for successive block pairs. The supplement gives the distributions and paired intervals. These measurements concern actual uniform maps on padded YCbCr 4:4:4; they motivate spatial allocation but are not cropped mixed-map losses or results for independently trained shallow codecs.

4.3. Early exit reduces work; routing improves its allocation

At the nominal 0.1 dB target, FLEX-UF with learned routing saves 27.97% of modelled synthesis MACs. Source-informed search reaches 29.52%, while uniform depth and dithering reach 21.07% and 25.43%, respectively (Fig. 4). The ordering separates three effects. Shallower synthesis supplies most of the reduction. Mixing adjacent depths improves the available cost resolution. Content-dependent placement contributes a further 2.54 percentage points over dithering, with a paired 95% interval of [1.83,3.26].

4.4. The gain persists under a common delivered-loss cap

Does the margin persist when every selected output meets the same measured loss cap? For each policy and frame-QP pair, we deduplicate the maps measured at the six nominal targets, retain those meeting the cropped Δ_{444} cap within 10^{-4} dB, and choose the cheapest. Repeated maps have identical recorded losses and costs. The evaluated e15 full-frame output supplies a zero-loss, zero-saving fallback; no quality point is interpolated.

Table 1 reports the 0.1 dB case. The router saves 27.75% versus 24.81% for dithering, a 2.93-point paired difference [2.11,3.79]. Their mean achieved losses are also close: 0.0786 and 0.0791 dB. Only one router case uses the full-frame fallback, compared with two dither cases. The margin

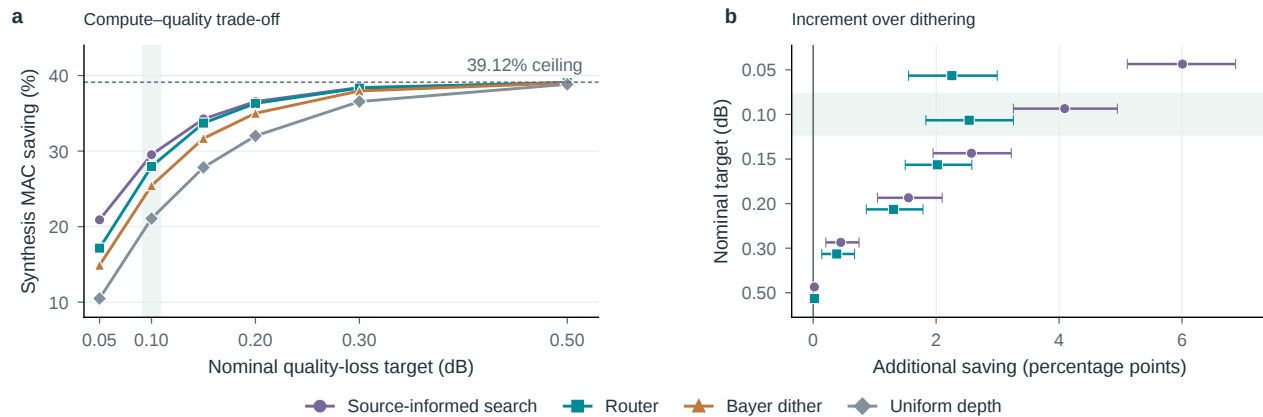


Figure 4. **Early exit supplies the saving; spatial routing adds a budget-dependent margin.** **a**, Modelled synthesis MAC reduction for four stopping rules. **b**, Paired advantages over Bayer dithering, with 95% sequence-cluster intervals; the shaded row highlights 0.1 dB. QP and fixed-cohort analyses are in the supplement. All curves use per-source calibration. MAC savings exclude router and signalling overhead and are not measured runtime reductions. Targets use the Δ_{444} MSE-ratio loss defined in Sec. 3.

Table 1. **Recorded outputs under a common 0.1 dB YCbCr 4:4:4-loss cap.** All 265 pairs are included. Loss is relative to cropped e15 full-frame synthesis. MAC saving excludes the retrospective candidate-selection cost.

Policy	MAC saved (%)	Mean loss (dB)	Fallbacks (/265)
Uniform depth	22.98	0.0670	11
Bayer dither	24.81	0.0791	2
Learned router	27.75	0.0786	1
Source search	29.19	0.0779	2

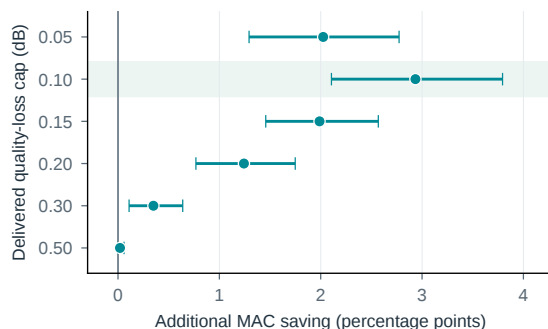


Figure 5. **Placement matters most before the cheap exit saturates.** Paired router–dither MAC-saving differences under a common measured YCbCr 4:4:4-loss cap, with 95% sequence-cluster intervals. All 265 pairs remain; e15 full-frame is the fallback. Selection is retrospective and source-aware.

survives a common measured cap. Selection remains retrospective and source-aware, omits candidate-acquisition cost, and searches only the recorded outputs.

4.5. Why the routing margin contracts

The cheapest reachable depth imposes a 39.12% ceiling on the arithmetic model. As the target relaxes, more tiles can take that exit and there are fewer consequential spatial decisions. The nominal router–dither margin declines from 2.54 points at 0.1 dB to 1.31 at 0.2 dB, 0.38 at 0.3 dB and 0.02 at 0.5 dB. Under the delivered-cap protocol the margin similarly falls from 2.93 to 0.35 points between 0.1 and 0.3 dB. The fixed-cohort result and QP breakdown in the supplement indicate that this behaviour is not explained solely by a changing feasible subset. At the 0.1 dB delivered cap, routing saves more in 170 pairs, less in 24, and ties in 71. At 0.5 dB, 262 of 265 pairs tie; the residual gain comes from one sequence.

The maps in Fig. 6 make the mechanism visible. At the

0.1 dB target, the router spends different depths on different regions. At 0.3 dB, search, routing and dithering all assign depth six on this frame, saving 39.1% MACs at 0.201 dB measured YCbCr 4:4:4 loss. We select this example from the five archived thumbnails by proximity to the aggregate 0.1 dB margin. Source luma supplies context, not a visual quality comparison.

A histogram-preserving diagnostic further separates spatial placement from exit proportions. Randomly permuting a map leaves its cost unchanged. We compute the exact expected table MSE under that permutation and compare it with the recorded assignment. At 0.1 dB, the router’s mean placement gain is 0.0213 dB [0.0167, 0.0262], whereas dithering’s is 0.0007 dB [−0.0002, 0.0017]. This table-based

Table 2. **Fixed controls trade average saving against quality risk.** Actual CPU reconstructions on all 53 QP32 frames. Controls are fitted to padded 444 loss on other sequences; the table reports cropped RGB loss against e15 full-frame. Equal calibration targets do not imply equal achieved quality. The final column counts losses above 0.1 dB.

Control	Policy	MAC saved (%)	RGB loss (dB)	> 0.1 (/53)
Mean	Uniform	19.78	0.0775	14
	Dither	23.80	0.0881	16
	Router	25.68	0.0868	17
Q90	Uniform	13.03	0.0516	2
	Dither	14.20	0.0564	2
	Router	16.74	0.0568	4

association holds the histogram fixed; replaying permuted maps through the final head remains a separate ablation.

4.6. Fixed controls test the routing advantage

Source-calibrated allocation gives the selector information that a cheap stopping rule would ideally avoid. We therefore fit its scalar control on four sequence folds and apply it to the fifth, keeping all QPs of each sequence together. Calibration minimises compute subject to a mean or 90th-percentile (Q90) table-loss target. We then replay every fixed map at QP32 through the actual network: 53 frames, three policies and two controls give 318 reconstructions, with no quality-based fallback.

Table 2 reports cropped RGB loss against e15 full-frame synthesis. Q90 calibration lowers the observed frequency of losses above 0.1 dB, at the cost of less arithmetic saving. The router–dither saving difference is 1.87 points under mean control, with a paired 95% interval of $[-1.60, 5.19]$, and 2.54 points under Q90, with interval $[-0.42, 5.54]$. Both include zero, and the policies do not have exactly matched achieved quality. Thus the source-calibrated margin has not yet been established for fixed controls.

This is a calibration diagnostic on development-used data, not an untouched test set. Two similar-content pairs cross the sequence folds; grouping them in a bootstrap sensitivity check leaves both intervals crossing zero. An aligned released-weight replay also measures the reference drift: e15 full-frame loses 0.00908 dB mean RGB PSNR against released D12. Using released D12 raises absolute policy losses by that same per-image offset without changing their paired differences. The supplement gives the split, predictor and metric audits.

4.7. Checking the mechanism

Exit alignment and repair. With the checkpoint and 53 Q90 maps fixed, we replace adapters, repair, or both

by identity and retain all images. Adapter removal costs 1.2309 dB mean RGB PSNR, with a paired 95% interval of $[0.8765, 1.6582]$. Repair removal costs only 0.00208 dB $[0.00060, 0.00363]$; removing both costs 1.2538 dB. The trained model depends strongly on exit alignment. The small repair benefit motivates a matched-retraining and timing comparison before increasing its capacity. These interventions measure dependence of fixed weights, not the best performance attainable without each module.

Stream-only reconstruction. A separate CPU decoder receives the common entropy payload and explicit two-bit exit indices. On two predeclared QP32 frames with four maps each, all eight images, latents, symbols and entropy indexes match encoder-side reconstruction exactly; inner entropy bytes stay identical across maps. This verifies the shared-payload path on those cases. Map/header bytes and malformed-stream checks are in the supplement; native CUDA interoperability and full-cohort timing remain unmeasured.

4.8. What the encoder pays for a stopping decision

Source-informed search needs candidate reconstructions to populate its error table. The archived QP32 benchmark records 1,023.26 ms for four distinct candidate outputs and 336.87 ms for a shared tiled pass. These upper medians pool 24 trials over eight frames; raw timing distributions and device load were not retained. They indicate the cost of acquiring decision information, not an end-to-end speedup. The supplement details the stages.

A fixed-control predictor can avoid this table, but the evaluated source-calibrated router cannot: its per-frame β uses the same error information. The next latency experiment must compare full decision paths, including feature generation, calibration, signalling, conditional execution and final verification. At a loose quality budget, where the router and blind maps nearly coincide, even small additional overhead may remove the practical benefit of content prediction.

5. From Shared Exits to Independent Depths

Shared early exit commits every region to one encoded representation. An independent codec can specialise its representation for a smaller synthesis network, but forfeits that reuse. This motivates our complementary 2/4/6/8/10/12-block DCVC-UF bank (Fig. 1), where a later ClassSR-inspired selector will choose a codec before encoding each region. D2/D4/D6 are training; D8/D10 are planned and D12 is the released anchor. No six-expert routing result is claimed.

A controlled capacity frontier. The scratch runs preserve the pinned official 105-epoch recipe [9], including

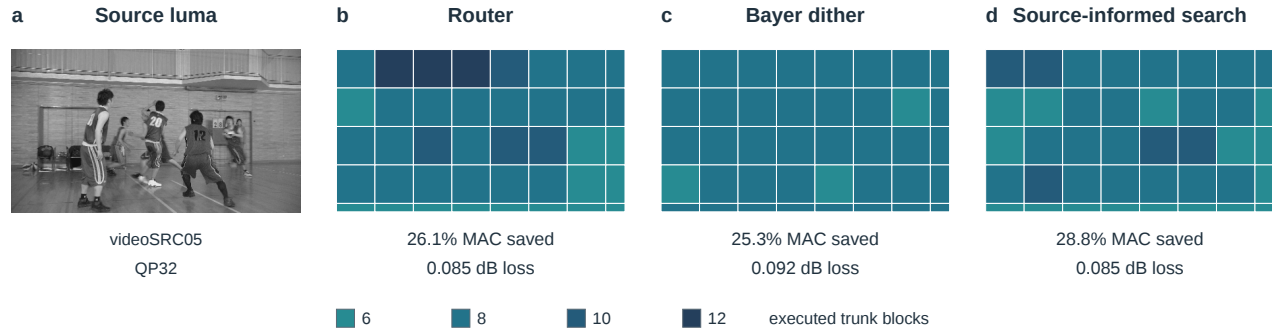


Figure 6. **Adaptive depth changes where synthesis work is spent.** **a**, Source luma for spatial context. **b–d**, Recorded router, Bayer and source-search maps at the 0.1 dB target, QP32. Colour gives exit depth; annotations give synthesis MAC saving and cropped YCbCr 4:4:4 loss. Uniform depth eight gives 24.2% saving and 0.082 dB loss. Maps use valid image extent; the text states the example-selection rule.

all 384,795 Open Images train_0/1/2 images, batch size 16 and FP32. Analysis and entropy weights learn alongside synthesis, so equal QP does not imply equal coded rate. Two fixed milestones, epochs 20 and 30, each evaluate 2,000 actual CPU bitstreams on 100 DIV2K centre crops across D2/D4/D6 and released D12. On the same 99 images at 0.2 payload bpp, D6’s RGB advantage over D2 changes from 0.183 to 0.107 dB; its epoch-30 difference from D4 is unresolved. Depth ordering is therefore not stable at these unfinished checkpoints. The supplement reports both milestones. Final same-recipe D12 and matched shared-exit training remain necessary to isolate depth and weight sharing.

The spatial cost of specialisation. Patchwise expert selection also changes geometry and entropy boundaries. With identical D6 weights, independently coded regions lose quality at matched rate; increasing context alone does not remove their rate cost. In a separate fixed D2/D6 assignment, merging compatible neighbouring regions improves matched-rate quality while retaining the same per-pixel depth map. These controls, detailed in the supplement, make patchwise D12 and blind expert mixtures essential baselines before a selector receives credit. The eventual comparison must match achieved rate and quality, charge signalling and grouping, and measure the effect of expert occupancy on latency. It will test when independent specialisation is worth more than shared-prefix reuse.

6. Discussion

Adaptation should pay for the decisions it changes. FLEX-UF identifies a budget-dependent role for spatial allocation. Under source calibration, tight quality targets leave room for useful placement. Near the shallowest-exit ceiling, a blind mixture reaches nearly the same arithmetic saving. This motivates a budget-conditioned choice between fixed, blind and learned policies, with its switching rule fixed on

validation data.

Reuse and specialisation are different commitments. Shared exits retain one representation and the work already performed; independent experts can adapt their encoders and synthesis capacity more freely. The bank adds model residency, entropy boundaries and occupancy-dependent execution costs. Its quality must be paired with those costs, not with shared-prefix arithmetic. Multiple exits inside each expert are outside the present design.

The remaining evidence gap. The source-calibrated study measures available allocation value, while the fixed-control replay leaves the routing increment unresolved. Content-grouped calibration on untouched images is the next test of that increment. Native bitstreams and paired wall-clock measurements must then include entropy work, prediction, tile movement, grouping and fallback. The current confidence intervals cover sample variation at fixed weights, not independent training seeds. Results are confined to intra reconstruction: YCbCr-MSE and RGB diagnostics do not establish perceptual equivalence, temporal consistency or performance of DCVC-UF’s chunk-based video path.

7. Conclusion

FLEX-UF makes synthesis depth a spatial decision inside DCVC-UF. Aligned exits and a shrinking tile batch reuse a common representation while omitting later block evaluations. Most measured arithmetic saving comes from cheaper depth; learned placement adds a smaller, budget-dependent increment. Blind mixtures and fixed-control replay separate these effects. Matched training and complete execution measurements will determine when independent-depth experts justify giving up shared computation.

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